Introduction to Knowledge Engineering

Assignment – CO1

Individual Assignment **Name: DINDU PUJITHA**

Duration: 1 Week **Reg No:192373037**

Total Marks: 50

CO1: Apply fundamental concepts of knowledge engineering such as knowledge representation, knowledge acquisition, rule-based reasoning, semantic networks, frames, ontologies and knowledge graphs to model, acquire and manage domain knowledge.

Topics covered: Knowledge Representation – Knowledge Acquisition – Knowledge-Based Systems – Expert Systems – Semantic Networks – Frames – Production Rules – Ontologies – Knowledge Graphs – Knowledge Management Systems.

Instructions: Q1, Q2, Q4, Q5, Q6 and Q7 ask you to build your own examples, so every submission should differ. Make sure every trace, calculation and table follows from your own data. Q3 uses the fixed rule base given in the paper. Diagrams may be hand-drawn or digital.

Q1 – Represent Domain Knowledge (Emergency Triage)	6 Marks

Problem Context – Hospital Triage. A hospital wants to capture the knowledge of experienced triage nurses to assign each arriving patient a priority of Critical, Urgent or Routine. The nurses use symptoms such as chest pain, blood pressure, body temperature and age.

Your Challenge

1. Identify at least three pieces of knowledge in the triage domain and classify each as declarative, procedural or heuristic knowledge.
2. Represent three of them in predicate logic.
3. Represent the same knowledge as production rules and as a frame describing a patient.
4. Evaluate logic, rules and frames against representational adequacy, inferential adequacy, inferential efficiency and acquisitional efficiency.
5. Identify two difficulties in representing triage knowledge (for example incomplete, uncertain or ambiguous knowledge).

Analysis Inquiry

Using the triage domain, explain the difference between data, information and knowledge.

Evidence Expected: *Classified knowledge items, logic and rule representations, a patient frame, and an evaluation of representation schemes.*

Answer: Represent Domain Knowledge (Emergency Triage)

1. Knowledge items and classification:

• Declarative knowledge: A body temperature above 38°C is generally treated as fever. This is a factual statement describing a condition in the triage domain.

• Procedural knowledge: The nurse assesses the patient in a systematic order: check airway, assess breathing, examine circulation and vital signs, and then record the findings. This describes how to perform the task.

• Heuristic knowledge: An older patient reporting chest pain together with high blood pressure may need critical priority. This is an experience-based guideline and must be applied with clinical judgement.

2. Predicate logic representations:

• $\forall x\ [(Patient(x) \wedge Temperature(x) > 38) \rightarrow Fever(x)]$

• $\forall x\ [(Patient(x) \wedge Age(x) > 50 \wedge ChestPain(x) \wedge HighBP(x)) \rightarrow Priority(x, Critical)]$

• $\forall x\ [(Patient(x) \wedge Fever(x) \wedge Age(x) < 5) \rightarrow Priority(x, Urgent)]$

The universal quantifier means that the rule applies to every individual who satisfies the stated conditions. Predicates describe the patient, symptoms and resulting triage category.

3. Production rules and patient frame:

R1: IF temperature is greater than 38°C THEN record Fever = Yes.

R2: IF age is greater than 50 AND chest pain is present AND blood pressure is high THEN set Priority = Critical.

R3: IF fever is present AND age is below 5 THEN set Priority = Urgent.

Patient frame: Patient_Ramesh {name: Ramesh; age: 58; chest_pain: Yes; blood_pressure: High; temperature: 37°C; priority: Critical}. The priority is inferred by R2. A real triage system must also consider additional clinical signs and approved hospital protocols.

4. Evaluation of representation schemes:

• Representational adequacy: Predicate logic represents precise facts and relationships; production rules represent practical IF–THEN knowledge; frames represent patient details through named slots and values.

• Inferential adequacy: Logic supports formal deduction; rules support forward and backward chaining; frames support inheritance and default values but usually need rules or procedures for complex conclusions.

• Inferential efficiency: Simple rule matching and frame-slot lookup are usually fast. General logical theorem proving may require more computation.

• Acquisitional efficiency: Nurses can often express their experience as rules, making rule acquisition convenient. Frames require a carefully designed structure, while predicate logic requires more formal knowledge of symbols and relationships.

5. Difficulties: First, symptoms may be incomplete when a patient is unconscious or unable to communicate. Second, terms such as severe pain or high blood pressure may be uncertain or depend on age and medical history. The system should record missing values, represent confidence where appropriate and allow a nurse to review uncertain cases.

Analysis inquiry – Data, information and knowledge: Data is a raw reading such as 150/95 mmHg. Information is the interpretation that this reading is high for the particular patient. Knowledge is the general clinical guideline that combines blood pressure with age and symptoms to help decide urgency. Data describes a measurement, information gives it meaning, and knowledge supports a decision.

Q2 – Acquire Knowledge from an Expert (Crop Disease Advisory)	**6 Marks**

Problem Context – Paddy Disease Advisory. A state agriculture department wants an advisory system that helps farmers identify paddy diseases. A plant-pathologist gave the following statement during a first interview:

Expert statement

"When paddy leaves show brown oval spots with grey centres and the weather has been humid for several days, I suspect blast. If the spots are mainly on the lower leaves and the farmer has used a lot of nitrogen fertiliser, I become more certain. But if the whole leaf is turning yellow with no spots, I would think of nutrient deficiency rather than a disease, and I would first ask for a soil test."

Your Challenge

1. Identify the roles in the project (domain expert, knowledge engineer, end user) and at least three sources of knowledge other than the expert.
2. Select three knowledge-elicitation techniques and justify each for this domain.

3. Prepare five interview questions that would follow up the statement above.
4. Convert the statement into at least four production rules and mark where certainty is involved.
5. Explain the knowledge-acquisition bottleneck, identify two difficulties in this project, and state how you would validate the acquired rules.

Analysis Inquiry

Why is tacit knowledge difficult to elicit, and which technique would help most in capturing it?

Evidence Expected: *Roles and sources, justified techniques, interview questions, rules derived from the transcript, and a validation plan.*

Answer : Acquire Knowledge from an Expert (Crop Disease Advisory)

1. Project roles and knowledge sources:

• Domain expert: the plant pathologist, who understands paddy diseases, symptoms, weather conditions and diagnosis.

• Knowledge engineer: interviews the expert, organises the knowledge, converts it into rules and checks that the system represents the expert's reasoning correctly.

• End user: farmers and agricultural extension officers who consult the advisory system.

Other sources include agricultural university research papers, government crop-protection manuals and bulletins, historical field diagnosis records, and the practical experience of extension officers.

2. Knowledge-elicitation techniques:

• Structured or semi-structured interviews help obtain explicit information about symptoms, weather, fertiliser use, diagnostic thresholds and exceptions.

• Observation with think-aloud protocol: the expert examines real plants or photographs and explains each judgement. This helps reveal visual clues and tacit knowledge that may not emerge in a normal interview.

• Card sorting: the expert groups photographs or symptom descriptions by likely disease. The groupings reveal how the expert distinguishes blast from other diseases.

3. Follow-up interview questions:

1) How many consecutive humid days make blast more likely, and does temperature affect the diagnosis?

2) What visual differences distinguish blast spots from brown spot and sheath blight?

3) What quantity of nitrogen fertiliser do you consider excessive for a given area of paddy?

4) Which symptoms or field conditions would make you reject blast as the likely diagnosis?

5) When a whole leaf turns yellow without spots, which nutrient deficiencies should be considered and what soil tests are recommended?

4. Production rules derived from the expert statement:

R1: IF leaves have brown oval spots with grey centres AND the weather has been humid for several days THEN suspect paddy blast (illustrative CF = 0.60).

R2: IF R1 conditions are present AND spots are mainly on lower leaves AND nitrogen use is high THEN increase confidence in blast (illustrative CF = 0.85).

R3: IF the whole leaf is yellow AND no spots are present THEN suspect nutrient deficiency rather than blast (illustrative CF = 0.70).

R4: IF nutrient deficiency is suspected THEN recommend a soil test before giving a specific nutrient treatment (procedural rule; CF = 1.00).

The certainty factors are example values for demonstrating uncertainty, not measured probabilities. They must be confirmed by the plant pathologist using field cases.

5. Knowledge-acquisition bottleneck and validation: The bottleneck occurs because expert knowledge is difficult and time-consuming to extract, organise and encode. One difficulty is that visual diagnosis is partly tacit and difficult to describe. Another is that experts may disagree about thresholds or symptom importance. Validate the rules against historical cases with known diagnoses, obtain an independent expert review, test the system on new field observations and record agreement, false diagnoses and cases requiring human review.

Analysis inquiry – Tacit knowledge: Tacit knowledge is experience-based skill that an expert may use without being able to explain every step. Observation and think-aloud protocol analysis is especially useful because it captures the expert while diagnosing actual leaves. The engineer can then ask follow-up questions and use card sorting to check whether the extracted visual cues are consistent.

Q3 – Reasoning with Production Rules (Loan Pre-screening)	8 Marks

Problem Context – Bank Loan Pre-screening. A bank encodes its pre-screening knowledge as the following rule base. Rules are tried in the order R1 to R6.

Rule	IF (conditions)	THEN (conclusion)
R1	Salaried AND Income_Stable	Employment_Secure
R2	Credit_Score_Good AND No_Defaults	Good_Credit
R3	Employment_Secure AND Good_Credit	Low_Risk
R4	Low_Risk AND Amount_Within_Limit	Approve_Loan

Rule	IF (conditions)	THEN (conclusion)
R5	Self_Employed AND Income_Stable AND Has_Collateral	Employment_Secure
R6	Low_Risk AND Amount_Exceeds_Limit	Refer_To_Manager

Case A facts: Salaried, Income_Stable, Credit_Score_Good, No_Defaults, Amount_Within_Limit.

Case B facts: Self_Employed, Income_Stable, Has_Collateral, Credit_Score_Good, No_Defaults, Amount_Exceeds_Limit.

Your Challenge

1. Apply forward chaining to Case A. Show the rule fired at each step and the facts added to working memory.
2. Apply backward chaining to Case A with the goal Approve_Loan. Show the goals and sub-goals.
3. Apply forward chaining to Case B. State whether Approve_Loan can be derived and what the system concludes.
4. Explain what backward chaining would do for the goal Approve_Loan in Case B, and which question the system would have to ask the user.
5. Describe two conflict-resolution strategies and state which one you used.
6. Explain how contradictory, redundant or circular rules would affect a rule-based system and how they can be detected.

Analysis Inquiry

When is forward chaining preferable to backward chaining, and when is backward chaining preferable?

Evidence Expected: *Step-by-step traces, goal tree, working-memory contents, and interpretation of the conclusions.*

Answer: Reasoning with Production Rules (Loan Pre-screening)

1. Case A – forward chaining:

Initial working memory: Salaried, Income_Stable, Credit_Score_Good, No_Defaults, Amount_Within_Limit.

Step 1: R1 fires because Salaried AND Income_Stable are true. Add Employment_Secure.

Step 2: R2 fires because Credit_Score_Good AND No_Defaults are true. Add Good_Credit.

Step 3: R3 fires because Employment_Secure AND Good_Credit are now true. Add Low_Risk.

Step 4: R4 fires because Low_Risk AND Amount_Within_Limit are true. Add Approve_Loan. Final conclusion: Case A qualifies for loan approval under the supplied rules.

2. Case A – backward chaining for Approve_Loan:

The goal Approve_Loan requires R4, so the system checks Low_Risk and Amount_Within_Limit. Amount_Within_Limit is already a fact. To prove Low_Risk, R3 requires Employment_Secure and Good_Credit. Employment_Secure is proved by R1 from Salaried and Income_Stable. Good_Credit is proved by R2 from Credit_Score_Good and No_Defaults. Both sub-goals are satisfied, therefore Low_Risk and then Approve_Loan are proved.

3. Case B – forward chaining:

Initial facts: Self_Employed, Income_Stable, Has_Collateral, Credit_Score_Good, No_Defaults, Amount_Exceeds_Limit.

Step 1: R2 fires and adds Good_Credit. R1 does not fire because Salaried is absent.

Step 2: R5 fires because Self_Employed, Income_Stable and Has_Collateral are true. Add Employment_Secure.

Step 3: R3 fires and adds Low_Risk.

Step 4: R6 fires because Low_Risk AND Amount_Exceeds_Limit are true. Add Refer_To_Manager. R4 cannot fire because Amount_Within_Limit is not established. Therefore Approve_Loan cannot be derived; the application is referred to a manager.

4. Case B – backward chaining: To prove Approve_Loan, R4 needs Low_Risk and Amount_Within_Limit. Low_Risk can be proved through R3, R5 and R2. Amount_Within_Limit is neither an initial fact nor the conclusion of any supplied rule. The system would ask, “Is the loan amount within the permitted limit?” The case states Amount_Exceeds_Limit, so the answer is No and the approval goal fails. The system can then establish Refer_To_Manager using R6.

5. Conflict-resolution strategies: Rule priority or rule order chooses the first applicable rule; specificity chooses the rule with the greatest number of matching conditions; recency gives preference to rules using newer facts. For these traces, use the stated order R1 to R6 and fire the first applicable rule that has not already fired (with refraction to avoid repeated firing).

6. Rule problems: Contradictory rules may derive incompatible conclusions from the same facts and make the result depend on rule priority. Redundant rules repeat the same conclusion and waste processing time. Circular rules can cause repeated inference or non-terminating backward searches. Detect them by reviewing rules, checking overlapping conditions and conclusions, building a dependency graph, and testing representative cases.

Analysis inquiry – Forward versus backward chaining: Forward chaining is data-driven and is useful when facts arrive continuously and the system must derive all possible outcomes, such as monitoring.

Backward chaining is goal-driven and is useful when a specific conclusion must be checked, such as whether a loan can be approved, because it investigates only the facts needed for that goal.

Q4 – Semantic Networks and Frames (Library Domain)	5 Marks

Problem Context – University Library. Consider the following description.

- Library members are either student members or faculty members. Every member has a name and membership ID, and by default may borrow 3 books for 14 days.
- Faculty members are an exception: they may borrow 10 books for 30 days.
- Books are either textbooks or reference books. Textbooks can be lent; reference books cannot be lent.
- Anil is a student member and Dr. Rao is a faculty member. 'Introduction to AI' is a textbook and 'Britannica' is a reference book. Anil has borrowed 'Introduction to AI'.

Your Challenge

1. Draw a semantic network showing is-a links, instance-of links and at least one other relation.
2. Represent the same knowledge as frames with slots, default values and inherited values.
3. Use inheritance to answer: How many books can Anil borrow? What is Dr. Rao's loan period? Can Anil borrow 'Britannica'?
4. Explain how a frame overrides a default value to handle an exception.
5. Compare semantic networks and frames and state one limitation of each.

Analysis Inquiry

What problem arises if a person is both a student member and a faculty member? Suggest how it can be resolved.

Evidence Expected: *Semantic network, frame hierarchy, inheritance reasoning, and comparison of the two schemes.*

Answer : Semantic Networks and Frames (Library Domain)

1. Semantic network (text representation): Member has Student_Member and Faculty_Member as subclasses (is-a links). Student_Member and Faculty_Member are kinds of Member. Anil is an instance of Student_Member; Dr_Rao is an instance of Faculty_Member. Book has Textbook and Reference_Book as subclasses. Introduction_to_AI is an instance of Textbook; Britannica is an

instance of Reference_Book. Anil —borrowed→ Introduction_to_AI is an additional relationship. Textbook —can_be_lent→ Yes and Reference_Book —can_be_lent→ No.

2. Frames and slots:

Member frame: name, membership_id, borrow_limit = 3 (default), loan_days = 14 (default).

Student_Member frame: is-a Member; inherits borrow_limit = 3 and loan_days = 14.

Faculty_Member frame: is-a Member; overrides borrow_limit = 10 and loan_days = 30.

Book frame: title and lendable status. Textbook frame: is-a Book; lendable = Yes. Reference_Book frame: is-a Book; lendable = No.

Anil frame: instance-of Student_Member; name = Anil; borrowed = Introduction_to_AI. Dr_Rao frame: instance-of Faculty_Member; name = Dr. Rao. Introduction_to_AI frame: instance-of Textbook. Britannica frame: instance-of Reference_Book.

3. Inheritance answers: Anil inherits the Member default and may borrow 3 books for 14 days. Dr. Rao belongs to Faculty_Member, so his loan period is 30 days and his borrowing limit is 10 books. Anil cannot borrow Britannica because Britannica is a Reference_Book whose lendable slot is No.

4. Default overriding: A frame first checks the instance slot and then searches its parent frames. Faculty_Member provides its own borrow_limit and loan_days values, so those values replace the general Member defaults for faculty. Other member types continue to inherit the defaults.

5. Comparison and limitations: A semantic network represents knowledge as nodes and labelled links and is easy to visualise, but link meanings may be informal and complex exceptions can be difficult to represent. Frames organise an object or concept into slots, defaults and inherited values, making structured descriptions convenient; however, large frame systems can become complicated and multiple inheritance may create conflicting slot values.

Analysis inquiry – Multiple membership: If one person is both a student and faculty member, the person may inherit conflicting borrowing limits and loan periods. Resolve this by defining an explicit class-precedence rule, creating a combined Student_Faculty frame with agreed values, or assigning local values to that person's instance. The policy should be chosen by the library rather than guessed by the system.

Q5 – Design an Expert System (Laptop Fault Diagnosis)	7 Marks

Problem Context – Laptop Fault Diagnosis. A service centre wants an expert system that helps a front-desk technician diagnose common laptop faults (power, overheating, slow performance) from symptoms reported by the customer.

Your Challenge

1. Draw the architecture of the expert system and state the role of each component in this application.

2. Write at least six production rules for the knowledge base, with a certainty factor (CF) for each.
3. **Show a sample consultation: facts supplied, rules fired and the final recommendation.**
4. Show the explanation the system would give for a WHY question and for a HOW question.
5. Show how certainty factors are combined when two rules support the same conclusion, with the calculation.
6. Compare an expert system with a conventional program and state one limitation of expert systems.

Analysis Inquiry

Why is the separation of the knowledge base from the inference engine important?

Evidence Expected: *Architecture diagram, rule base, consultation trace, explanation facility examples, and certainty-factor calculation.*

Answer : Design an Expert System (Laptop Fault Diagnosis)

1. Architecture and component roles: User/technician → User Interface → Inference Engine ↔ Working Memory; the Inference Engine consults the Knowledge Base and sends results to the Explanation Facility. A Knowledge Acquisition Module allows experts to update the Knowledge Base. The user interface collects symptoms and displays advice; working memory stores the current case; the inference engine matches and fires rules; the knowledge base stores diagnostic rules; the explanation facility describes WHY and HOW; the acquisition module supports rule maintenance.

2. Production rules with illustrative certainty factors:

R1: IF laptop does not power on AND charger LED is off THEN suspect faulty charger (CF 0.80).

R2: IF laptop does not power on AND charger LED is on AND battery drains quickly THEN suspect faulty battery (CF 0.70).

R3: IF laptop overheats AND fan noise is high THEN suspect dust in fan (CF 0.90).

R4: IF laptop overheats AND fan is not spinning THEN suspect faulty fan (CF 0.85).

R5: IF laptop is slow AND overheating is present THEN suspect dust in fan (CF 0.50).

R6: IF laptop is slow AND disk usage is consistently high THEN suspect disk or software issue (CF 0.60).

R7: IF dust in fan is suspected THEN recommend cleaning the fan and heat sink (CF 1.00).

These CF values are illustrative strengths of expert belief and should be calibrated using service-centre cases.

3. Sample consultation: The technician records Overheating (CF 0.9), Fan_Noise_High (CF 0.8), and Slow_Performance (CF 0.7). Fan_Not_Spinning and High_Disk_Usage are No. R3 derives Dust_In_Fan with CF = 0.90 × min(0.9, 0.8) = 0.72. R5 also derives Dust_In_Fan with CF = 0.50 ×

min(0.7, 0.9) = 0.35. Combining these positive supports gives 0.72 + 0.35 × (1 − 0.72) = 0.818. R7 then recommends cleaning the fan and heat sink with CF 0.818, approximately 0.82. R4 and R6 do not fire because their required symptoms are absent.

4. Explanation facility: WHY question – "Why are you asking whether the fan noise is high?" Answer: "Because rule R3 uses high fan noise together with overheating to support the conclusion that dust may be present in the fan." HOW question – "How did you recommend cleaning the fan?" Answer: "R3 and R5 both supported Dust_In_Fan. Their certainty factors were combined to 0.818, and R7 uses Dust_In_Fan to recommend cleaning."

5. Certainty-factor combination: For a rule, conclusion CF = rule CF × minimum certainty of its premises. Thus R3 = 0.9 × min(0.9, 0.8) = 0.72, and R5 = 0.5 × min(0.7, 0.9) = 0.35. When two positive rules support the same conclusion, CFcombined = CF1 + CF2(1 − CF1) = 0.72 + 0.35(0.28) = 0.818. This method assumes the evidence is suitable for combination and the CF values are meaningful.

6. Expert system versus conventional program: An expert system stores domain knowledge explicitly as rules and uses an inference engine to reason with facts; a conventional program generally follows procedures and algorithms written directly in code. Expert systems can explain which rules led to a conclusion and are useful for heuristic problems. Conventional programs are often better for fixed calculations and well-defined algorithms. A limitation of expert systems is that they cannot reliably diagnose faults outside their encoded knowledge and depend on experts to keep rules accurate and current.

Analysis inquiry – Separation of knowledge base and inference engine: Separating them allows experts to change or add diagnostic rules without rewriting the inference engine. The same engine can be reused with a different knowledge base, and rules can be reviewed and tested independently. It also makes explanations easier because the system can show which stored rules produced a recommendation.

Q6 – Ontologies and Knowledge Graphs (University Domain)	**5 Marks**

Problem Context – University Knowledge Graph. A university wants to link information about students, professors, courses and departments so that questions can be answered across its separate databases.

Your Challenge

1. Design a small ontology with at least five classes, a class hierarchy, and at least four properties with domain and range.
2. Write at least eight facts as subject–predicate–object triples.

3. Draw the resulting knowledge graph.
4. **Write and execute a graph query: "Which students are enrolled in courses taught by a named professor?"**
5. Show one inference that follows from the class hierarchy or from a property's domain or range.
6. Compare a taxonomy, an ontology and a knowledge graph.

Analysis Inquiry

How do ontologies support knowledge sharing and reuse? State one limitation.

Evidence Expected: *Ontology definition, triples, graph diagram, query trace, inference example and comparison.*

Answer : Ontologies and Knowledge Graphs (University Domain)

1. Ontology classes and hierarchy: Person is a general class. Student and Professor are subclasses of Person. Course and Department are additional classes. This gives five classes: Person, Student, Professor, Course and Department.

Properties with domain and range: teaches (domain Professor, range Course); enrolledIn (domain Student, range Course); worksIn (domain Professor, range Department); offeredBy (domain Course, range Department). These definitions specify the expected type of subject and object for each relationship.

2. Example subject–predicate–object triples:

(Dr_Meena, type, Professor); (Dr_Meena, worksIn, CSE); (Dr_Meena, teaches, ML101); (ML101, offeredBy, CSE); (Arun, type, Student); (Arun, enrolledIn, ML101); (Dr_Kumar, type, Professor); (Dr_Kumar, teaches, DB102); (Priya, type, Student); (Priya, enrolledIn, DB102); (DB102, offeredBy, CSE). Each triple connects one subject to an object through a named predicate.

3. Knowledge graph (text diagram): Dr_Meena —type→ Professor; Dr_Meena —worksIn→ CSE; Dr_Meena —teaches→ ML101; ML101 —offeredBy→ CSE; Arun —type→ Student; Arun —enrolledIn→ ML101. Similarly, Dr_Kumar —teaches→ DB102 and Priya —enrolledIn→ DB102. Nodes represent entities or classes and arrows represent relationships.

4. Graph query: { ?student enrolledIn ?course . Dr_Meena teaches ?course }. First, Dr_Meena teaches ML101, so ?course = ML101. Next, Arun enrolledIn ML101, so ?student = Arun. The answer is Arun. Priya is not returned because she is enrolled in DB102, which Dr_Meena does not teach.

5. Inference: Student is a subclass of Person. Since Arun has type Student, the system can infer that Arun is also a Person even if the Person triple is not explicitly stored. Similarly, the domain of teaches

is Professor, so a subject known to teach a course can be inferred to be a Professor under the ontology's modelling assumptions.

6. Comparison: A taxonomy is mainly a hierarchy of terms, such as Student is-a Person. An ontology defines classes, properties, constraints and formal meanings, such as the domain and range of teaches. A knowledge graph stores individual entities and their relationships, such as Dr_Meena teaches ML101. An ontology can provide the schema and semantics used to organise and reason over a knowledge graph.

Analysis inquiry – Sharing and reuse: An ontology provides a common vocabulary and formal definitions, allowing different university systems to interpret Student, Course and Department consistently. This supports integration and reuse of information across admissions, learning management and staff databases. One limitation is the effort required to build, agree on and maintain the ontology; inconsistent definitions can reduce interoperability.

Q7 – Diagnose and Improve a Knowledge Management System	**13 Marks**

Problem Context – IT Help-Desk Knowledge Base. A college keeps solutions to common IT problems (Wi-Fi, passwords, VPN, printing, software, exam portal) as articles in a searchable knowledge base. Students type a question and the system returns the best-matching article. After a semester, many students complain that the system returns no answer, an irrelevant answer or an outdated article. The faculty suspects limited or duplicated knowledge, an outdated or conflicting article, a flat and unstructured representation, poor search, or missing knowledge that only the support staff carry in their heads.

A. Diagnose the Knowledge Problem — 2 Marks

- Identify possible causes of the poor answers.
- Distinguish knowledge-quality problems (content) from system-design limitations (representation and retrieval).

Evidence Expected: *A diagnosis supported by example queries and the answers returned.*

Answer : Diagnose the Knowledge Problem

Possible content-quality causes include outdated VPN instructions, duplicate articles, missing solutions, inaccurate steps and missing terms used by students. For example, the old VPN article A4 (2019) competes with the current A5 (2025). Staff may know that "eduroam" refers to campus Wi-Fi but that knowledge has not been documented.

System-design causes include flat keyword-only representation, no synonym handling, no concept tags, no article status filter and tie-breaking by article order. These limitations can return an irrelevant or retired article even when a correct current article exists.

Example diagnosis: “internet not working in hostel” may return no result because the article uses “wifi”; “unable to sign in to my account” may fail because the article says “login”; “how to use VPN from home” may select the older VPN article if both articles match equally. Content problems require review and updating of articles, while retrieval problems require better representation, indexing and ranking.

B. Define the Knowledge Objective — 2 Marks

- Define the knowledge to be managed, its users and its representation.
- Specify a measurable objective for improving answer quality.

Evidence Expected: *A clearly defined task and a measurable objective.*

Define the Knowledge Objective

The knowledge to be managed is the college’s collection of verified solutions for Wi-Fi, passwords, VPN, printing, software and exam-portal issues. The main users are students; support staff and faculty members act as content contributors and reviewers. Each article should include a title, problem description, solution steps, concept tags, synonyms, owner, last-reviewed date and status.

The task is to return the most relevant current help article for a student’s free-text question. A measurable objective is to achieve at least 80% top-one accuracy on a fixed test set, return zero retired articles, and leave no more than 10% of queries unanswered.

C. Select and Redesign the Method — 3 Marks

- Choose an approach, for example keyword search, rule-based troubleshooting, ontology/concept tagging, or knowledge graph.
- Justify the choice against the data and users, and propose a modification that addresses the limitation.

Evidence Expected: *Method selection, justification, and a revised knowledge-management workflow or diagram.*

Select and Redesign the Method

I select a lightweight ontology-based concept-tagging method with a synonym dictionary and article-status control. This suits a small help-desk collection because the main problems are vocabulary differences and outdated duplicate articles; a full knowledge graph or complex expert system would be unnecessary for this initial version.

Redesign steps: (1) review articles and mark outdated duplicates as retired; (2) assign each current article one or more concepts such as Network, Account_Access, Printing or Software; (3) create

synonyms such as internet/wifi/network and sign-in/login/account; (4) convert query words into concepts; (5) rank only current articles by number of matched concepts and specificity; (6) show the selected article and allow students to report an incorrect or missing answer.

Workflow: Capture student questions and staff knowledge → Validate article accuracy → Organise articles with concepts, synonyms and status → Retrieve the best current match → Collect feedback and unanswered terms → Review and update the knowledge base.

D. Predict the Outcome — 1 Mark

- Predict how the change will affect the answers returned before you test it.

Evidence Expected: *A clear, testable prediction.*

Predict the Outcome

Before testing, I predict that concept tags and synonyms will improve retrieval for questions that use different words from the articles. Retiring the old VPN article should prevent the system from returning outdated instructions. I expect top-one accuracy to increase from the keyword baseline, no retired article to be returned, and the "eduroam" query may remain unanswered until that synonym is added.

E. Implement and Evaluate — 3 Marks

- Implement the original and revised methods using Python or a clearly traceable manual procedure.
- Evaluate both on the same set of test queries, report the results and identify remaining errors.

Evidence Expected: *Code or traceable calculations, a comparison table and evaluation results.*

Implement and Evaluate

The evaluation should use the same fixed set of ten queries for both the original keyword method and the revised concept-tagged method. For each query, record the expected article, the article returned by each method, whether the result is correct, whether it is unanswered and whether it is retired. Accuracy is calculated as correct top-one results divided by total queries × 100.

Illustrative traceable test set and results (these are example results for the proposed small dataset, not evidence of a live college deployment):

Query: "forgot my password" → expected A1; keyword A1; revised A1 (correct).

Query: "internet not working in hostel" → expected A2; keyword no result; revised A2 (correct).

Query: "wifi not working" → expected A2; keyword A2; revised A2 (correct).

Query: "how to use VPN from home" → expected A5; keyword may return retired A4; revised A5 (correct).

Query: "login to exam portal error" → expected A8; keyword A8; revised A8 (correct).

Query: "unable to sign in to my account" → expected A1; keyword no result; revised A1 (correct).

Query: “print my assignment in lab” → expected A6; keyword A6; revised A6 (correct).
Query: “install MATLAB on laptop” → expected A3; keyword A3; revised A3 (correct).
Query: “lost student card” → expected A7; keyword A7; revised A7 (correct).
Query: “eduroam keeps dropping” → expected A2; keyword no result; revised no result if eduroam is not yet in the synonym dictionary.
Illustrative comparison: keyword baseline = 6/10 correct (60%), 3 unanswered and 1 retired article returned. Revised concept-tagged method = 9/10 correct (90%), 1 unanswered and 0 retired articles returned. The remaining error is the missing “eduroam” synonym. These figures should be replaced with actual execution output if the implementation is run on a real help-desk dataset.

F. Justify the Final Design — 2 Marks

- Decide whether the objective was achieved and explain the limitations of the final system.
- State when another approach would be preferable and how the knowledge will be kept up to date.

Evidence Expected: *An evidence-based conclusion and a justified recommendation.*

Justify the Final Design

For the illustrative test above, the objective is achieved: accuracy increases from 60% to 90%, no retired article is returned, and only 10% of queries remain unanswered. The result supports the prediction that concept tagging, synonyms and article-status filtering improve retrieval.
The design still has limitations. The dataset and test set are small, a synonym dictionary needs regular maintenance, and a query may contain spelling errors, multiple issues or unfamiliar wording. The test queries should be separated into development and held-out evaluation sets to avoid overstating accuracy.
A statistical or embedding-based search method may be preferable when the collection becomes very large or students use varied natural-language questions. A rule-based troubleshooting system is preferable when the student must answer a sequence of diagnostic questions. Keep knowledge current by assigning an owner and review date to every article, collecting feedback, reviewing unanswered searches and retiring obsolete instructions after verification.